

1. Mechanical properties affected by atomic structure

- Which mechanical properties (e.g., strength, ductility, toughness) are most sensitive to atomic arrangement?

One of the most important aspects in this regard is slip systems. These are a combination of a slipping plane (plane with highest atomic density) and a slipping direction (direction with highest atomic density). These define the crystallographic directions along which slipping (motion of dislocations) occurs most easily. In general FCC is the most ductile, followed by BCC and lastly HCP is the least ductile. This pretty much lines up with the number of slip systems (FCC having 12, BCC having 48 but a low packing density means dislocations are harder to move, and HCP has 3-6).

In terms of yield strength and plastic deformability the closely packed planes of FCC allow more dislocation motion and therefore a lower yield strength whereas a more sparse packing such as HCP results in a stronger but more brittle material.

- What role do atomic packing and defects play in determining material performance?

Defects play a very important role in terms of material performance. First of all, all plastic deformation is a result of defect motion. This means that many defects also results in an easier time plastically deforming a material.

A low atomic packing density and a lot of defects will also oftentimes result in the material being a lot weaker as a lot of the internal structure suddenly is removed.

- How do crystal bonding types (metallic, covalent, ionic) influence mechanical behavior?

In general a high bonding energy (e.g. metallic bonds) can be thought of as the atoms being connected by very tight springs. This results in a high strength as a lot of force is required to deform the material but the very stiff springs also risk breaking if pulled too tightly, which corresponds to a more brittle material. Vice versa is true for a weaker e.g. covalent bond.

2. Deformation behavior at room temperature – FCC vs BCC metals

- How do slip systems differ between FCC and BCC structures?

A slip system is a combination of a slip plane (highest atomic density plane) and a slip direction (highest atomic density direction). Together this slip plane corresponds to the plane along which dislocations move the most easily.

As a rule of thumb, more slip systems means that dislocations have an easier time moving and therefore the material becoming more ductile. However, for FCC (having 12 slip systems) and BCC (having ~48) this is not the case. Here FCC is actually more ductile than BCC. This happens as the slip planes of the BCC systems are not as closely packed and they therefore require more energy to move.

- Why are FCC metals generally more ductile at low temperatures?

As the FCC metals have 12 closely packed slip systems as opposed to the 48 non-close packed slip systems of BCC less thermal energy is required for dislocation motion (the atoms simply do not have to move as far in the FCC structure as in the BCC).

- How does temperature influence dislocation mobility in BCC metals?

BCC metals do not have closely packed planes. This leads to a high Peierls stress (the stress required to move a dislocation). As the temperature increases so does the thermal energy (which is actually just kinetic energy) of the atoms. This increase in kinetic energy leads to dislocations having an easier time overcoming the energy barriers required for movement.

3. Brittleness at low temperatures

- How does the ductile-to-brittle transition temperature relate to crystal structure?

The ductile-to-brittle transition temperature is a name for the temperature at which a metal apparently undergoes a sudden ductile-to-brittle transition. As the high Peierls stress in BCC metals means that dislocations can get stuck at low temperatures these metals often have a DBTT. In FCC metals the dislocations have an easier time moving due to the higher atomic packing density meaning that these metals often do not have a DBTT.

- What materials are known for brittle fracture at low temperatures, and why?

Especially metals with a BCC structure such as steel are infamous for their DBTT. This is the temperature at which dislocations get stuck hindering further dislocation motion and rendering the material brittle.

- How does impact testing help identify temperature-sensitive brittle behavior?

If one conducts a series of impact tests on a material at different temperatures a temperature vs. absorbed energy plot can be made. On this it should become clear that below a certain temperature the metal stops absorbing energy and becomes brittle rather than ductile.

4. Same elastic modulus, different grain sizes

- How does grain size affect yield strength and why?

A large grain size implies fewer grain boundaries and the opposite is true for a smaller grain size. If a material has a lot of grain boundaries dislocations have a hard time moving through the material and they can effectively get stuck rather easily. This is governed by the Hall-Petch relation:

$$\sigma_y = \sigma_0 + \frac{k}{\sqrt{d}}$$

Where σ_y is the yield strength, σ_0 is the “intrinsic” yield strength and d is the average grain diameter. K is a proportionality constant.

- What does the Hall-Petch relationship suggest about fine vs coarse grains?

As a rule of thumb (if your grains are not too big nor too small) the Hall-Petch relation describes how yield strength is impacted by grain size. It sounds:

$$\sigma_y = \sigma_0 + \frac{k}{\sqrt{d}}$$

Where σ_y is the yield strength, σ_0 is the “intrinsic” yield strength and d is the average grain diameter. K is a proportionality constant.

- In what applications would fine-grained metals be preferable despite similar stiffness?

A finer grain size with a similar stiffness means that the strength and toughness is increased. This would be preferable in high-load applications such as car engines.

5. Stress–strain curve and necking behavior

- What key points define a stress–strain curve (yield, UTS, fracture)?

A stress-strain curve typically starts by going linearly upwards. This corresponds to the region of elastic deformation. As one passes the yield strength the stress-strain curve will flatten out. This is the region of plastic deformation. At the ultimate tensile strength the graph will top. This is where necking starts. In actuality the stress actually is constantly increasing as the cross sectional area of the specimen shrinks faster than the stress, however, the engineering stress will start to fall which is why the stress appears to fall on the stress-strain curve. A short time hereafter the material will typically fracture.

- What does necking indicate about the material’s ability to sustain load?

Necking is a sign of a ductile material. Necking typically is preferable so long as it does not hinder the strength too much – after all it is better that your specimen necks than that it fractures in a brittle manner. However necking is typically not something one hopes for as a necked material is severely hindered in its mechanical performance.

- How does true stress–strain behavior differ post-necking?

In actuality after necking sets in, the stress does not decrease as the cross sectional area of the specimen will shrink. The reason that the stress-strain curve appears to decrease in magnitude after necking is simply put a result of one using the engineering strain when calculating stress-strain and the engineering strain does not take this shrinkage into consideration.

6. Strain hardening mechanism

- What microscopic changes cause strain hardening in metals?

Strain hardening is the process of plastically deforming a metal at cold temperatures (hence the alternative name cold working). The plastic deformation one induces under strain hardening leads to a motion of dislocations and it also introduces new dislocations. As the amount of dislocations increases the dislocations can “trap” each other impeding further movement. On a stress strain

curve this process will become apparent as “shifting” the x-axis in a way leading to a higher yield strength.

- How is strain hardening represented on a stress–strain curve?

After reaching the yield strength of a material the stress-strain curve stops being linear. Instead plastic deformation (strain hardening) happens. When the strain hardening is stopped the stress-strain curve will go downwards linearly. In this way the yield strength will increase as the distance from the stress strain curve to the x-axis is higher after plastic deformation.

- Why does the material become stronger yet less ductile?

It can be explained by the stress strain curve by the above mechanism. Simply put the elastic part of the stress-strain curve is shifted along the x-axis – this leads to a shorter amount of ductility before breakage.

Microscopically as one introduces more dislocations and thus impedes further dislocation motion it makes sense that trying to force the dislocations to move across each other will result in breakage.

7. Elastic modulus and strengthening

- Why does the elastic modulus remain largely unchanged by dislocation interactions?

The elastic modulus mainly depends on the interatomic interaction between the atoms in the material. A high level of interaction (strong bonds/tight springs) leads to a stiff material and v.v.

On an atomic basis the interatomic forces are by far overpowering any external force that one might try to apply to the material under normal conditions. This means that these interatomic interactions pretty much do not care about what you do to the material.

- What atomic-level property defines the modulus?

The interatomic interactions (i.e. the bond strength) strong bonds gives a stiff material and vice versa.

- How does this explain the predictability of stiffness across metal classes?

Because the chemical groupings (metal classes) are based on electron properties of the atoms. These electron properties are responsible for the interatomic interaction and thus the stiffness.

8. FCC, BCC, HCP structures and mechanical properties

- What distinguishes these crystal structures in terms of atom arrangement?

FCC – Face centered cubic. A unit cell consists of an atom at the center of each face. This leads to 6 half-atoms and 8 quarter atoms = 4 total atoms per unit cell.

BCC – Body centered cubic. A unit cell consists of an atom at the center of the unit cell. This leads to 1 whole atoms and 8 quarter atoms = 2 atoms per unit cell.

HCP – Hexagonal close packed. A unit cell is a hexagonal prism. Each of the 12 hexagonal corner contains $\frac{1}{6}$ of an atom. 2 half atoms at each face and 3 whole atoms in the middle of the prism = 6 atoms per unit cell.

- How do differences in slip systems affect ductility?

Slip systems are a combination of a slip direction and a slip plane. Together these define the direction along which dislocations are most likely to travel (i.e. the only road they will travel). In this way a high number of slip systems leads to a lot of ways the dislocations can travel – this increases ductility.

However, a high number of slip systems is not the only factor when it comes to ductility. How easily these slip systems are activated also plays an important role. This mainly comes down to how closely packed the atoms are as closer packing leads to less energy required for an atom to move. For FCC having 12 slip systems that are all closely packed this leads to lots of ductility. BCC has about 48 slip systems however none of these are as closely packed as the FCC systems are. This means that the activation energy (high Peierls stress) for moving atoms is higher leading to less ductility. The HCP has about 3-6 systems that are all rather hard to activate leading to very brittle materials.

- Which common metals fall into each category?

FCC: Ductile metals such as aluminum, lead and gold are excellent examples.

BCC: Strong and hard but brittle metals such as steel and chromium fall into this category. These often display prominent ductile-to-brittle-transitions.

HCP: These metals typically display limited ductility even at elevated temperatures. Here magnesium and titanium are great examples.

9. Deformation at 70% of melting temperature

- At high homologous temperatures, what deformation mechanism dominates?

Especially creep is prominent at high temperatures. This is the process at which a material will change its dimensions (strain) if a constant stress is applied. This is different from normal elasticity/plasticity as with creep you are applying an often relatively low stress but you are doing it for a prolonged time.

- How can you assess creep risk in design?

As creep is a slow process that normally takes a lot of years you normally do not have time for testing creep under normal conditions. Therefore tests are conducted at higher temperatures for shorter time from which data can be extrapolated. To do this one normally employs the Larson-Miller parameter that quantifies the change in creep given a change in temperature.

- What testing methods evaluate high-temperature deformation?

A common test used to assess creep rate is subjecting a specimen to a fixed stress at a high temperature for an amount of time. The creep can then be plotted against the time giving a creep curve.

10. Creep at $>0.4T_m$

- What are the stages of creep and their microstructural signatures?

Creep is normally split into three stages:

1. Primary creep: The initial phase. Here the creep rate decreases as it starts out creeping a lot before “settling in”. In this phase the dislocations start by gliding along all the most favourable slip systems and the specimen slowly starts to strain harden as the dislocations tangle. This corresponds a higher dislocation density and more tangling of these on the microstructural plane.
 2. Secondary creep: Here the creep rate is constant producing a linear creep curve. Here a few different mechanisms are at play, but most importantly there has been found a balance between the amount of new dislocations produced and the amount recovered mainly by diffusion etc. Here the grains are approximately equiaxed and boxed in by dislocations walls.
 3. Tertiary creep: Here the creep rate starts increasing before rupture happens. Here voids start forming at grain boundaries which slowly coalesce into microcracks and so forth.
- How do dislocation and grain boundary mechanisms evolve through creep stages?

In the primary phase the dislocations are free to move rather undisturbed through the slip planes. As the dislocations start to tangle and the second phase is entered the dislocations mainly move by climbing out of their slip plane and into a new one by diffusion. This process sometimes leads to dislocation annihilation as a dislocation climbs into the plane it was supposed to be in meaning the dislocation ceases to exist. In the tertiary phase the dislocation motion is rather localized around voids leading to crack formation.

- Why is secondary (steady-state) creep most critical in design?

Simply put it is the phase that lasts the longest – the material spends very little time in phase 1 and 3 and thus the nice predictable creep of the secondary phase becomes what one designs for.

11. Solute strengthening and atomic size

- How do solute atoms hinder dislocation motion?

Solute atoms introduce lattice strain, especially interstitial diffusion strains the lattice quite a bit. Dislocations also carry around lattice strain. As nature likes to minimize energy the solute atoms and the dislocations will organize themselves in a way that minimizes energy and strain by partially cancelling each other out.

- Why does atomic size mismatch matter?

For substitutional solute atoms the size mismatch is the main driving factor behind introducing the aforementioned lattice strains.

- How is substitutional vs interstitial strengthening different?

Interstitial solute atoms sit between the spaces in the normal crystal lattice. This normally strains the lattice quite a bit more than is the case for substitutional solutions. This increased lattice strain leads to more dislocations being “pinned” thus hindering their motion.

12. Grain boundary strengthening and high-temperature behavior

- Why do grain boundaries become less effective with temperature?

Grain boundaries normally hinder dislocation motion and diffusion. As the temperature rises these processes also accelerate as more atoms surpass the energy barrier.

- What role does diffusion play in weakening boundaries?

Diffusion is one of the main mechanisms behind weakening grain boundaries. Especially at high temperatures. As more atoms surpass the energy barrier diffusion from one grain to another leads to grain boundaries weaken.

- How does grain growth impact strength at elevated temperatures?

At elevated temperatures (above the recrystallization temperature) the grains will start to grow in size (annealing). As the grains grow the density of grain boundaries decrease thus removing some of the obstacles for dislocation motion.

13. Point defects in crystalline solids

- What are vacancies and interstitials, and how do they form?

A vacancy is a “hole” in the crystal lattice where an atom is “supposed” to be but is not. Interstitials are atoms that have “lodged” themselves into the crystal lattice. These form mainly due to random chance i.e. temperature induced motion leads to some atoms jumping out of their normal spot. Interstitials rarely happens with the same atom as the rest of the lattice as this introduces enormous strains. Instead a small foreign atom such as carbon or chloride is typically used.

- How do these defects influence electrical conductivity?

A high amount of defects leads to a high electrical resistance. Electricity is carried by electrons and these are scattered by anything that is not the perfectly ordered crystal lattice.

- What is the role of point defects in diffusion?

Point defects enable diffusion. This is because the energy required for diffusing into a vacancy is a lot smaller than the energy required to diffuse into the place another atom is thus forcing it away. Interstitial diffusion typically happens a lot faster as the energy barrier is a lot smaller.

14. Dislocation motion and resistance

- What does an edge and screw dislocation look like?

An edge dislocation is an extra half plane of atoms that terminates within the crystal. A screw dislocation does not introduce a new group of atoms but instead an entire half of the face of the crystal has shifted a bit compared to the face opposite.

- How does slip occur along preferred crystallographic planes?

Slip planes are planes with the highest atomic density. This high atomic density leads to dislocations having an easier time moving as they do not have to move as far.

- What impedes dislocation movement (e.g., solutes, grain boundaries)?

Anything that is not the standard normal crystal lattice impedes dislocation movement in some way. This is because any configuration except the perfect crystal has lattice strains that can pin dislocations.

15. Grain size, Hall–Petch, and other strengthening mechanisms

- How does the Hall–Petch equation relate grain size to yield strength?

The Hall-Petch equation says that the increase in tensile strength is proportional to the inverse of the square root of the diameter as:

$$\sigma \propto d^{-\frac{1}{2}}$$

- What are two alternative methods for increasing metal strength?

On top of reducing grain size one can also choose to use solid solution strengthening, where solute atoms are introduced in the crystal lattice and thereby introducing lattice strain. Cold working is another method. Here the amount of dislocations is increased thus enabling the dislocations to get tangled.

- Why is there a limit to grain refinement's effectiveness?

Because as the amount of grain boundaries increase the grain boundaries become the dominant feature of the microstructure. As this happens the grains can start to slide against each other thereby causing cavitation. This can actually end up weakening the material in a process known as the inverse Hall-Petch effect.

16. FCC lattice and packing factor

- How do you construct an FCC unit cell diagram?

Face centered cubic. Contains 6 half atoms at the center of each face and 8 quarter atoms at the corners. This corresponds to a total of 4 atoms per unit cell in total.

- How is the atomic packing factor calculated?

The atomic packing factor is an idealized way to think of the packing density. Here one imagines the atoms as hard spheres and line them up in the unit cell such that they just touch. Then one simply finds the quotient of the volume of these atoms and the total volume of the unit cell.

- Why does FCC have high ductility?

FCC possesses high ductility as it has rather many slip planes (12) and these are all close packed. This means the atoms do not have to move as far to slip (i.e. do not have to expend as much energy) and they are therefore more likely to slip leading to plastic deformation and thus ductility.

17. Fatigue and S–N curves

- How does cyclic loading cause failure below yield stress?

Due to fatigue. Fatigue is a description of the phenomenon that even a small amount of stress will cause a material to fail if it is subjected to it enough times. This is true also if the force is less than the yield strength of the material and the material therefore should be able to withstand it. The mechanism behind this is thought to come down to cracks acting as stress amplifiers, meaning a stress will be greatly amplified at the tips of small cracks. In theory this can lead to very high localized stresses that can cause these cracks to grow. As the cracks grow the force is multiplied even more and so forth until the specimen breaks due to crack formation.

- What does a fatigue limit mean for steel vs aluminum?

The fatigue limit is a description of the fact that for some materials (such as steel) fatigue does not appear to accumulate for stresses below a certain threshold (the fatigue limit). Aluminum, and other metals that do not have a fatigue limit, on the other hand will fatigue even with a theoretical infinitesimal force applied – just a tiny bit.

I recently read a paper however talking about how they had found that steel failed even for stresses lower than its theoretical fatigue limit with very high numbers of cycles. This could challenge the validity of the fatigue limit as the idealized concept it is as of right now where one states that even an infinite amount of cycles below the fatigue limit will not cause failure.

- How are S–N curves generated and interpreted?

S-N curves are curves typically made during a fatigue test. These plot the stress intensity on the y-axis and the number of cycles until failure for a given specimen on the x-axis. These cycles are typically cycles in a rotating bending test where a specimen is fixed at both ends and rotated by a motor. At the same time a force is applied to the middle of the specimen and due to the specimen rotating this force will cause alternating localized compression and tension in the material.

18. Stress concentrations and fatigue life

- What is the effect of notches or scratches on local stress?

Notches and scratches act as localized stress amplifiers. This can be thought of as the force being spread over a smaller area leading to a higher localized stress although this might not be a technically accurate explanation. This is also why sharp edges on cracks make them more damaging as this amplifies the stress more.

- How does fracture initiation differ from propagation?

Fracture initiation is when small microvoids start to form inside the material. These small voids will slowly grow larger until they coalesce into proper cracks. Once a critical level of void coalescence has been reached the crack will rapidly propagate through the material and fracture.

- What tools or tests can quantify stress concentration effects?

I guess a fatigue test might help shine light on it although it is not really testing stress concentration magnitude. One can calculate a stress concentration factor k_t as:

$$K_t = \frac{\sigma_m}{\sigma_0}$$

19. Brittle vs ductile fracture

- What are the visual signs of each fracture type on the surface?

A brittle fracture is characterized by totally flat fracture surface. This is because little to no plastic deformation happens before fracture. A moderately ductile fracture has a characteristic cup and cone fracture surface and a very ductile material has a fracture surface kind of like play doh.

- How does stress state (triaxial vs uniaxial) influence fracture mode?

A uniaxial test has a specimen e.g. a bar that is only fixed in the ends. This allows for dislocation flow and poisson effects (changing of the cross section) leading to a more ductile fracture. A triaxial test has the specimen fixed at more locations and thus hinders dislocation movement more. This favors brittle fracture.

- What prevention methods are used in brittle materials?

The main brittle failure prevention mechanisms revolve around removing or at least minimizing the harmful stress amplifying effects of cracks and sharp edges.

20. Fracture toughness vs strength

- How is fracture toughness defined and measured?

The fracture toughness is a measurement of the resistance of a material to brittle failure when a crack is present. This is given by:

$$K_{Ic} = Y\sigma_c\sqrt{\pi a}$$

Where Y is a dimensionless factor which accounts for the geometry of the specimen, σ_c is the critical stress for crack propagation and a is the crack length. Before this stress intensity is reached cracks will not propagate rapidly through the material.

- Why does strength not fully predict failure risk?

There is always some inherent variability in the performance of different specimens of the same material. This is due to compositional and geometrical differences.

- How is fracture toughness applied in failure prediction?

If one applies a stress in excess of the fracture toughness the material will splinter. Therefore one can use the fracture toughness to find out how much one can load a specimen given its internal structure.

21. Eutectic alloys and cooling

- What is the eutectic point and why is it significant?

The eutectic point is a given composition and temperature for an alloy at which a liquid phase (containing both or all metals in the alloy) solidifies into two distinct solid phases upon cooling – this is not the normal case as a mixture of two liquids typically will solidify at different temperatures and that one of the solid phases therefore will be formed totally before the other starts forming. This is nice because working with a eutectic alloy you are effectively working with a material that behaves like a proper “pure” metal upon solidification.

- What microstructures form upon slow cooling through the eutectic?

Due to the two phases forming simultaneously leading to a lamellar configuration of differing phases within the material. In non eutectic cooling one of the materials will solidify fully before the other starts meaning the two phases will not be intermixed.

- How do you identify the eutectic composition on a phase diagram?

You simply follow the liquidus line (the line separating the liquid mixed phase from the solid phases underneath) and identify its lowest point. This lowest point (corresponding to the temperature and composition at which the alloy can be a liquid at the lowest temperature) is the eutectic point.

22. Lever rule in binary diagrams

- What does the lever rule calculate in a two-phase region?

In the two-phase region multiple solid phases coexist, e.g. an α and a β phase. By using tie lines and the lever rules one is able to find the respective compositions of each of the coexistent phases and determine the phase fraction of each. This is done by first of all choosing an overall alloy

composition c_0 . At a given temperature one can then find the fraction of each phase in the two phase region by drawing an isotherm horizontally across the phase diagram. The concentration at which this tie line intersects the line separating the phase on the left hand side and the two phase region is the composition of that (e.g. the composition of the α grains c_α) and the concentration at which the tie line intersects the line separating the two phase region to the right hand side gives the composition of that (e.g. the composition of the β grains c_β). The lever rule then states that:

$$\text{Fraction of } \alpha = \frac{c_\beta - c_0}{c_\beta - c_\alpha}$$

$$\text{Fraction of } \beta = \frac{c_\alpha - c_0}{c_\alpha - c_\beta}$$

- How does it relate to mass conservation?

The lever rule is a direct result of mass conservation. It simply is a rearrangement of this:

$$c_0 = c_\alpha \cdot x_\alpha + c_\beta \cdot x_\beta$$

Which is a simple equation stating that the total composition should be the mass-weighted average of the compositions of the phases.

- In which regions of the phase diagram is the lever rule applicable?

Only in two-phase regions. In one phase regions the composition of the phase is trivial to find given the overall composition and in three-phase regions the lever-rule is not directly applicable but a similar reasoning can be followed from mass conservation as was the case above.

23. Binary phase diagram – eutectic system

- What are the key features of a binary eutectic diagram?

The most important feature of a eutectic binary diagram is the eutectic point with a fixed temperature and composition. This is the lowest point of the (often v-shaped) liquidus line (i.e. the lowest temperature at which the binary system can be a liquid). From the eutectic point and horizontally outwards a solidus line separates the liquid + solid regions ($L + \alpha$ or $L + \beta$) from the totally solid ones (α , β or $\alpha + \beta$). Each side of the diagram contain a solid phase field (α or β). These are both shaped a bit like flat triangles with the longest side oriented against either edge of the x-axis. Below the solidus line and between the solid single phase regions a solid solution region exists i.e. $\alpha + \beta$. This is bounded by solvus lines against the solid single phase regions at the sides.

- How do invariant reactions manifest on a cooling curve?

As a straight horizontal line on a curve of temperature vs. time. This could be due to e.g. solidification or vaporization.

- What microstructures result from rapid vs slow cooling?

As a eutectic binary system is cooled from the L region at a given overall composition it will at some point cross the liquidus line. Here some of the alloy will start turning solid. Unless you are right at the eutectic this will at first become a mixture of the liquid and one solid phase. The compositions of these will be different as the composition of the liquid phase will follow the liquidus line and the composition of the solid phase will follow that of the solidus line. This follows onwards with the same logic through the solvus line until room temperature is reached.

For slow cooling proper time is given for atomic processes such as diffusion to happen – this leads to large and uniform eutectic lamellae. For quick cooling not as much time is given for atomic diffusion. This leads to smaller grains and a higher chance of having uneven saturation across the material.

24. Eutectoid reaction in steel

- What is the eutectoid composition and temperature in steel?

0,77% C and 727 Celsius.

- What are the phases involved in the eutectoid transformation?

This is a eutectoid and not a “proper” eutectic reaction as in this case a solid phase (γ ; austenite, FCC of C in γ -Fe) transforms into two other solid phases; the BCC α -phase called ferrite which only has limited solubility of carbon and a cementite (or iron carbide / Fe_3C) phase is formed.

- How does the eutectoid differ from a eutectic?

In the eutectic a liquid solidifies into a two solid phases and in the eutectoid a solid phase transforms into two other solid phases.

25. Martensite formation and tempering

- What characterizes the martensitic transformation?

The martensitic transformation is famously the reason why quenching works. Here a piece of hot steel is cooled quickly. This transforms the steel from the FCC austenite with high carbon solubility to Martensite. Normally cold steel would transform to pearlite or bainite if cooled slowly. However when quenching the metal is cooled so quickly that the carbon that is dissolved in the FCC interstitial sites does not have time to move away and therefore gets trapped in position. This causes huge internal stresses and therefore a high strength.

- How does the lack of diffusion influence structure and hardness?

The lack of diffusion that is the reason quenching works is important because without no lattice strain would be induced. The lattice strain that results from the carbon atoms not having time to diffuse away is what gives Martensite its strengths.

- What does tempering do to microstructure and properties?

Tempering is the process of heating the quenched metal to below its eutectoid point for an extended amount of time before allowing it to cool. This adds energy to the atoms in the crystal lattice which helps the carbon atoms to overcome the forces pinning them in place allowing them to relax the strain on the structure a bit depending on the amount of time and temperature at which the heat treatment is undertaken.

26. Pearlite, bainite, martensite comparison

- How do cooling rates influence which microstructure forms?

By cooling faster, you allow less time for carbon to diffuse out of the system. In the slow cooled case one gets the nice lamellar structure of the pearlite. This is a eutectic mixture of α + Fe₃C. If you cool the metal moderately quickly then both α and Fe₃C will still be formed. They will however not have the time to organize themselves into the proper lamellae from pearlite. If you cool the metal quickly i.e. quenching, you get martensite. Here the carbon atoms do not have time to diffuse out of the crystal lattice and martensite is therefore produced.

- Compare the morphology and hardness of each phase.

Pearlite is characterized by a lamellar structure with either fine or coarse grains of these lamellar and alternating structure. In bainite not enough time is provided for this lamellar structure to emerge and instead the two phases coexist in a fine feathery structure that is rather messy. This leads to finer grains and it is therefore harder than pearlite. In martensite not enough time is allowed for the carbon to even diffuse out. Martensite is very hard as a lot of lattice stress is introduced.

- In what applications is each microstructure preferred?

As the cooling rate increases so does the hardness and at the same time the ductility decreases. For applications where some ductility is needed such as e.g. structural elements the more ductile pearlite could be preferred whereas martensite might be preferred for hard cutting bits etc.

27. Steel transformations upon cooling

- What are the nucleation and growth characteristics of pearlite and bainite?

At high temperatures (slow cooling) pearlite is formed rather homogeneously at the grain boundaries of the austenite. As more time passes more and more pearlite is formed until all austenite is gone. Bainite on the other hand is cooled more quickly. This results in the bainite not being formed at the edges of the austenite grain boundaries but instead spaced more randomly throughout the grains. These start off by nucleating without the carbon diffusing like for martensite but the temperature is in the case of bainite held just high enough that the carbon still is able to diffuse out after this “martensite” intermediary has been formed.

- How does martensite form differently from diffusional products?

For martensite the steel is cooled so quickly that the carbon does not have sufficient time to diffuse out. This distorts the crystal lattice heavily as the FCC austenite tries to change into BCC pearlite. The carbon atoms being trapped in here however distort this crystal lattice just a bit such that the BCC structure actually becomes a BCT structure.

- What is the significance of tempered martensite?

Martensite is although very hard often not a really useful material as it is extremely brittle. By tempering the metal the grains are given sufficient time and energy to help mitigate some of the internal stresses improving ductility.

28. TTT and CCT diagrams

- What is the purpose of TTT and CCT diagrams?

A TTT diagram (Time Temperature Transformation) is a diagram with time on the x-axis and temperature on the y-axis. These are isothermal in nature i.e. they tell you what phases form when if austenite is cooled to a fixed temperature and held there.

CCT diagrams (Continuous Cooling transformation) on the other hand represent a continuous cooling. These are often superimposed on TTT diagrams where they tell about what phases will form depending on the cooling rate.

- How do you use them to design heat treatments?

Because these basically give you the “recipe” for your heat treatment. I.e. you have a need for your steel to be able to do this or this, then you find out what that corresponds to in terms of microstructure whereafter the CCT and TTT diagrams are recipes for success.

- What are the key differences in their assumptions and usage?

TTT assumes isothermal “cooling” whereas CCT’s assumes a constant cooling rate.

29. Short fiber reinforcement and critical fiber length

- What determines the effectiveness of load transfer in short fibers?

The forces between the interface between the fibers and the matrix and the fibers. The stronger the fibers are bound the less likely they are to get pulled out. On top of this the length of the fibers should surpass that of the critical fiber length as this is a requirement for effective force transfer.

- How is the critical length derived?

I won't go into the full derivation but the critical fiber length basically is a measurement of whether or not a fiber is sufficiently long for the matrix to be able to deliver a force equal to the tensile strength of the fiber or not – i.e. is the fiber sufficiently long that its strength becomes the weak point and not the strength of the interface between the fiber and the matrix.

- What happens if fibers are below or above the critical length?

Fibers below the critical length are not able to take up as much stress from the matrix as they would be able to if they were longer.

30. **Thermoplastics vs thermosets**

- What structural features distinguish these two polymer types?

Thermoplastics have linear or branched chains without much cross linking. The chains are bound to each other by van der Waals forces and can therefore easily be melted by heat. Thermosets on the other hand have extensive cross linked structures. This makes it very hard for these to melt and therefore they will end up burning and not melting when sufficient energy is applied.

- How do processing methods differ (e.g., molding, recycling)?

Thermoplastics come as small granules that are heated hence they melt and this can then easily be molded – often many times. For thermosets these typically are made by combining a resin with a hardener, after this one has some time to form the product before it solidifies permanently. Think epoxy or rubber.

- What are typical use-cases for each based on mechanical properties?

Thermoplastics are typically softer than the thermosets however they are typically also better for the environment as they can be recycled.